

Test 38: Curie point of the vortex-flux lattice magnet (v23 three-pass)

Verdict: PASS | Generated: 2026-09-15 17:16:11 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

STATISTICAL+EXACT (v22.2 three-pass rewrite): order parameter $\chi(T)=\ln[p_{\text{GUE}}(s)/p_{\text{GOE}}(s)]?$? the KL lean ? on Chebyshev-unfolded spacings of a paired-realization vortex-

METHOD: STATISTICAL+EXACT (v22.2 three-pass rewrite): order parameter $\chi(T)=\ln[p_{\text{GUE}}(s)/p_{\text{GOE}}(s)]?$? the KL lean ? on Chebyshev-unfolded spacings of a paired-realiza

Result

Test 38 (v23 three-pass) t38: PASS • Curie 3-pass (72?+96?+38c) ? 486.9 min, 346 spectra in cache

‘plots/t38_curie_panels.png’ ? Test 38 ? Curie point (38a + 38b + 38c) (journal-engine multi-panel figure)

Plots

Reproduce with: julia ab_cloud_v23.jl --test 38 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(no log_comp entries)

FULL COMPUTATION LOG

Test 38 Curie ? lattice 38a: 72×72 (5184 nodes), hardcore 38b: 96×96 (9216 nodes), seeds 2

cache: 0 spectra loaded from ./t38_cache/spectra_v22C1.csv

CONSOLE CAPTURE

38c C2 staircase: L = 24 ? 36 ? 48 ? 64 ? 80 ? 96 • 6 rungs • 10 seeds/point

V1 • MC ?_GUE = +0.04755 – 0.00109 (quad +0.04743)

V1 • MC ?_GOE = -0.06607 – 0.00172 (quad -0.06427)

V1 • KS p: GUE 0.665, GOE 0.395 (n=20k)

V1 • ?r? MC/quad: GUE 0.6349/0.6368, GOE 0.5696/0.5709

38a coarse L=72: 14% (2/14) • 00:45 elapsed

38a coarse L=72: 21% (3/14) • 01:33 elapsed

38a coarse L=72: 36% (5/14) • 03:03 elapsed

38a coarse L=72: 43% (6/14) • 03:47 elapsed

38a coarse L=72: 50% (7/14) • 04:31 elapsed

38a coarse L=72: 64% (9/14) • 06:10 elapsed

38a coarse L=72: 71% (10/14) • 06:59 elapsed

38a coarse L=72: 86% (12/14) • 08:36 elapsed

38a coarse L=72: 93% (13/14) • 09:22 elapsed

38a coarse L=72: 100% (14/14) • 10:11 elapsed

38a coarse grid: done

38a refine L=72: 17% (2/12) • 00:46 elapsed

38a refine L=72: 25% (3/12) • 01:33 elapsed

38a refine L=72: 33% (4/12) • 02:23 elapsed

38a refine L=72: 42% (5/12) • 03:12 elapsed

38a refine L=72: 50% (6/12) • 04:02 elapsed

38a refine L=72: 67% (8/12) • 05:34 elapsed

38a refine L=72: 75% (9/12) • 06:20 elapsed

38a refine L=72: 83% (10/12) • 07:10 elapsed

38a refine L=72: 92% (11/12) • 07:56 elapsed

38a refine L=72: 100% (12/12) • 08:42 elapsed

38a refine grid: done (12 new spectra)

V1 • PASS

V2 • PASS

V3 • PASS

V4 • PASS

V5 • PASS

38a: $\chi^2(0)=+0.0339-0.0047$ $\chi^2(T_{\text{hi}})=-0.0686-0.0024$ depth=0.1025 $T^*(72)=1.7519$

38b endpoints L=96: 25% (1/4) • 00:00 elapsed

38b endpoints L=96: 50% (2/4) • 04:45 elapsed

38b endpoints L=96: 75% (3/4) • 09:31 elapsed

38b endpoints L=96: 100% (4/4) • 14:14 elapsed

38b refine L=96: 100% (9/9) • 39:10 elapsed

38b hardcore: done

38b controls: 25% (1/4) • 00:00 elapsed

38b controls: 50% (2/4) • 00:54 elapsed

38b controls: 75% (3/4) • 01:44 elapsed

38b controls: 100% (4/4) • 02:58 elapsed

A1 • PASS

A2 • PASS

A3 • PASS

A4 • PASS

A5 • PASS

A6 • PASS

A7 • PASS

38b: $T^*(96)=1.6473$ $\chi^2 T^*=-0.1046$ $\chi^2(0)=+0.0394$ depth=0.1238 $\chi^2=-0.951$

controls: no-flux $\tau(0)=-0.0734$ (want < 0.02) • static deph $\tau(0)=+0.0316$ (want > 0.02)

38c C1 density scan: 10% (1/10) • 00:00 elapsed

38c C1 density scan: 20% (2/10) • 00:59 elapsed

38c C1 density scan: 30% (3/10) • 02:05 elapsed

38c C1 density scan: 40% (4/10) • 03:34 elapsed

38c C1 density scan: 50% (5/10) • 04:35 elapsed

38c C1 density scan: 60% (6/10) • 05:31 elapsed

38c C1 density scan: 70% (7/10) • 06:29 elapsed

38c C1 density scan: 80% (8/10) • 07:30 elapsed

38c C1 density scan: 90% (9/10) • 08:31 elapsed

38c C1 density scan: 100% (10/10) • 09:20 elapsed

38c C1 density scan: done

C1 • PASS • robust f-range [0.50, 2.00], width 1.50

38c C2 FSS staircase L=24?36?48?64?80?96 (10 seeds/point)

C2 rung 1/6 L=24: 10% (5/50) • 00:02 elapsed

C2 rung 1/6 L=24: 20% (10/50) • 00:06 elapsed

C2 rung 1/6 L=24: 30% (15/50) • 00:09 elapsed

C2 rung 1/6 L=24: 40% (20/50) • 00:13 elapsed

C2 rung 1/6 L=24: 50% (25/50) • 00:16 elapsed

C2 rung 1/6 L=24: 60% (30/50) • 00:20 elapsed

C2 rung 1/6 L=24: 70% (35/50) • 00:23 elapsed

C2 rung 1/6 L=24: 80% (40/50) • 00:26 elapsed

C2 rung 1/6 L=24: 90% (45/50) • 00:30 elapsed

C2 rung 1/6 L=24: 100% (50/50) • 00:33 elapsed

C2 rung 2/6 L=36: 10% (5/50) • 00:06 elapsed

C2 rung 2/6 L=36: 20% (10/50) • 00:14 elapsed

C2 rung 2/6 L=36: 30% (15/50) • 00:21 elapsed

C2 rung 2/6 L=36: 40% (20/50) • 00:28 elapsed

C2 rung 2/6 L=36: 50% (25/50) • 00:35 elapsed

C2 rung 2/6 L=36: 60% (30/50) • 00:43 elapsed

C2 rung 2/6 L=36: 70% (35/50) • 00:50 elapsed

C2 rung 2/6 L=36: 80% (40/50) • 00:57 elapsed

C2 rung 2/6 L=36: 90% (45/50) • 01:05 elapsed

C2 rung 2/6 L=36: 100% (50/50) • 01:12 elapsed

C2 rung 3/6 L=48: 10% (5/50) • 00:21 elapsed

C2 rung 3/6 L=48: 20% (10/50) • 00:49 elapsed

C2 rung 3/6 L=48: 30% (15/50) • 01:20 elapsed

C2 rung 3/6 L=48: 40% (20/50) • 01:49 elapsed

C2 rung 3/6 L=48: 50% (25/50) • 02:19 elapsed

C2 rung 3/6 L=48: 60% (30/50) • 02:49 elapsed

C2 rung 3/6 L=48: 70% (35/50) • 03:16 elapsed

C2 rung 3/6 L=48: 80% (40/50) • 03:42 elapsed

C2 rung 3/6 L=48: 90% (45/50) • 04:17 elapsed

C2 rung 3/6 L=48: 100% (50/50) • 04:43 elapsed

C2 rung 4/6 L=64: 10% (5/50) • 01:55 elapsed

C2 rung 4/6 L=64: 20% (10/50) • 04:19 elapsed

C2 rung 4/6 L=64: 30% (15/50) • 07:12 elapsed

C2 rung 4/6 L=64: 40% (20/50) • 09:31 elapsed

C2 rung 4/6 L=64: 50% (25/50) • 11:33 elapsed

C2 rung 4/6 L=64: 60% (30/50) • 13:35 elapsed

C2 rung 4/6 L=64: 70% (35/50) • 15:50 elapsed

C2 rung 4/6 L=64: 80% (40/50) • 17:57 elapsed

C2 rung 4/6 L=64: 90% (45/50) • 20:06 elapsed

C2 rung 4/6 L=64: 100% (50/50) • 22:12 elapsed

C2 rung 5/6 L=80: 10% (5/50) • 06:04 elapsed

C2 rung 5/6 L=80: 20% (10/50) • 13:36 elapsed

C2 rung 5/6 L=80: 30% (15/50) • 22:05 elapsed

C2 rung 5/6 L=80: 40% (20/50) • 57:06 elapsed

C2 rung 5/6 L=80: 50% (25/50) • 1:04:46 elapsed

C2 rung 5/6 L=80: 60% (30/50) • 1:13:28 elapsed

C2 rung 5/6 L=80: 70% (35/50) • 1:25:25 elapsed

C2 rung 5/6 L=80: 80% (40/50) • 1:33:59 elapsed

C2 rung 5/6 L=80: 90% (45/50) • 1:44:02 elapsed

C2 rung 5/6 L=80: 100% (50/50) • 1:53:24 elapsed

C2 rung 6/6 L=96: 10% (5/50) • 23:43 elapsed

C2 rung 6/6 L=96: 20% (10/50) • 50:24 elapsed

C2 rung 6/6 L=96: 30% (15/50) • 1:11:40 elapsed

C2 rung 6/6 L=96: 40% (20/50) • 1:40:14 elapsed

C2 rung 6/6 L=96: 50% (25/50) • 2:01:46 elapsed

C2 rung 6/6 L=96: 60% (30/50) • 2:27:46 elapsed

C2 rung 6/6 L=96: 70% (35/50) • 2:56:01 elapsed

C2 rung 6/6 L=96: 80% (40/50) • 3:26:01 elapsed

C2 rung 6/6 L=96: 90% (45/50) • 3:51:36 elapsed

C2 rung 6/6 L=96: 100% (50/50) • 4:18:20 elapsed

38c C2 staircase: done

C2 • PASS • $m^*(T^*)$ stairs L=24/36/48/64/80/96 [-0.858, -0.438, -0.348, -0.320, -0.291, -0.371] vs anchor -0.160 • rising 4/5

C2 • Binder U? reported, not gated (n=10 realizations/point): L24 [+0.52 +0.48 +0.61 +0.67 +0.67] L36 [+0.40 +0.04 +0.29 +0.55 +0.61] L48 [+0.43 +0.22 +0.09 +0.54 +0.64] L64 [+0.42 +0.27 +0.46 +0.52 +0.59] L80 [+0.44 +0.27 +0.46 +0.59 +0.62]

C3 • disclosure rendered (8 lines, figure panel 6)

C4 • PASS • $\sup(\text{refine } 38a)=2.083$, $\sup(\text{refine } 38b)=2.002$? 3.0 = V4 bound

C5 • PASS • terminology enforced; C2 tie: critical trend consistent

[PASS] t38 Test 38 Curie: 38a PASS • 38b PASS • 38c PASS • $T^*(72)=1.752$ • $T^*(96)=1.647$ • 486.9 min

figure ? results/run_20260914_234626/test_38_curie_point/plots/t38_curie_panels.png (131.9 KB)

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